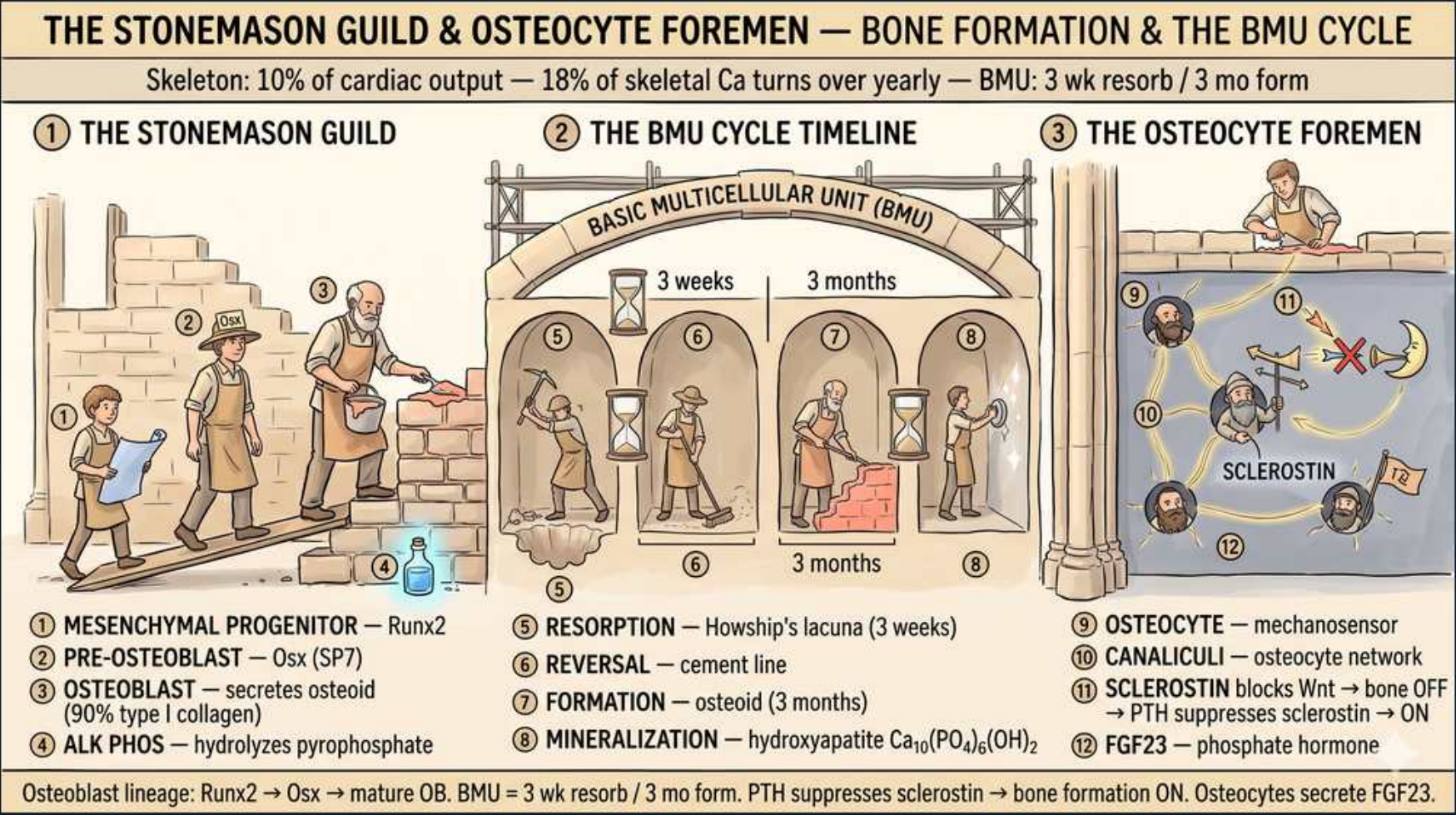




1. Mesenchymal progenitor → Runx2

WHAT YOU SEE

Stonemason apprentice with Runx2 blueprint

WHAT IT ENCODES

Osteoblasts derive from MESENCHYMAL stem cells, NOT hematopoietic. Runx2 is the master osteoblast transcription factor, followed by Osterix (Osx/SP7). Lineage: Runx2 → Osx → mature osteoblast. This is the opposite lineage from osteoclasts.

BOARD TRAP

OsteoCLAST comes from the macrophage/monocyte (hematopoietic) lineage — do not confuse the two; only osteoBLASTs are mesenchymal.

2. Pre-osteoblast → Osx (SP7)

WHAT YOU SEE

Worker mid-stage in the guild line

WHAT IT ENCODES

Osterix (SP7) acts downstream of Runx2 to commit pre-osteoblasts to the mature bone-forming phenotype. Required for osteoblast maturation and matrix deposition.

BOARD TRAP

Loss of Runx2 → cleidocranial dysplasia (absent clavicles, delayed fontanelle closure) — a classic board image.

3. Osteoblast secretes osteoid

WHAT YOU SEE

Mason laying mortar/osteoid

WHAT IT ENCODES

Mature osteoblasts secrete OSTEOID — unmineralized matrix that is ~90% type I collagen. Osteoid mineralizes ~10 days later into hydroxyapatite. Osteoblasts also express RANKL and OPG to control osteoclasts.

BOARD TRAP

Unmineralized osteoid accumulation = OSTEOMALACIA (adults) / rickets (children), seen in vitamin D deficiency — NOT osteoporosis (where mineralized bone mass is low).

4. Alkaline phosphatase

WHAT YOU SEE

ALK PHOS labeled tool/vial

WHAT IT ENCODES

Osteoblast alkaline phosphatase hydrolyzes pyrophosphate (a mineralization inhibitor), enabling hydroxyapatite formation. Serum ALP is a bone-FORMATION marker; it rises in Paget disease, osteomalacia, healing fracture, and bone mets.

BOARD TRAP

In primary hyperPTH ALP can be high too; but a markedly high ALP with normal Ca/PO4 in an older patient points to PAGET, not hyperparathyroidism.

5. Resorption — Howship's lacuna (3 wks)

WHAT YOU SEE

Arched BMU bay marked resorption

WHAT IT ENCODES

The BMU cycle starts with osteoclastic RESORPTION carving Howship's lacunae, lasting ~3 weeks. This is the brief phase — coupling means formation always follows.

BOARD TRAP

Antiresorptives (bisphosphonates, denosumab) reduce resorption depth; teriparatide instead stimulates the FORMATION phase — opposite mechanisms, both raise BMD.

6. Reversal — cement line

WHAT YOU SEE

Transition bay, cement line drawn

WHAT IT ENCODES

After resorption, the REVERSAL phase deposits a cement line; mononuclear cells prepare the surface and recruit osteoblasts (coupling). The cement line marks the boundary of each remodeling unit.

BOARD TRAP

Mosaic pattern of irregular cement lines = PAGET disease histology on the board.

7. Formation — osteoid (3 mo)

WHAT YOU SEE

Masons rebuilding the arch

WHAT IT ENCODES

FORMATION lasts ~3 months — far longer than the 3-week resorption phase. Net: BMU = 3 weeks resorb / 3 months form. This mismatch is why antiresorptives take months to raise BMD.

BOARD TRAP

Because formation lags resorption, in high-turnover states bone LOSS dominates — relevant to early estrogen-deficiency osteoporosis.

8. Mineralization — hydroxyapatite

WHAT YOU SEE

Finished mineralized arch

WHAT IT ENCODES

Osteoid mineralizes into hydroxyapatite $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$. Requires adequate Ca, phosphate, and alkaline phosphatase activity. Defective mineralization = osteomalacia.

BOARD TRAP

FGF23 excess (tumor-induced osteomalacia, XLH) causes phosphate wasting → impaired mineralization despite normal Ca — a non-vitamin-D osteomalacia.

9. Osteocyte — mechanosensor

WHAT YOU SEE

Foreman figure embedded in bone

WHAT IT ENCODES

Osteocytes are terminally differentiated osteoblasts buried in lacunae; they are the bone's MECHANOSENSORS and orchestrate remodeling. Most abundant bone cell. They secrete sclerostin and FGF23.

BOARD TRAP

Mechanical loading SUPPRESSES sclerostin → MORE bone; disuse/immobilization raises sclerostin → bone loss (immobilization hypercalcemia/osteoporosis).

10. Canaliculi — osteocyte network

WHAT YOU SEE

Channels linking osteocytes

WHAT IT ENCODES

Canaliculi connect osteocyte processes into a syncytial network for nutrient transfer and mechanotransduction signaling across the bone matrix.

BOARD TRAP

Loss of this network (microdamage) triggers targeted remodeling — relevant to atypical femoral fractures with over-suppressed turnover on long-term bisphosphonates.

11. Sclerostin — blocks Wnt → bone OFF

WHAT YOU SEE

SCLEROSTIN banner with blocked X

WHAT IT ENCODES

Sclerostin (gene SOST), secreted by osteocytes, inhibits Wnt/ β -catenin signaling → turns bone formation OFF. PTH SUPPRESSES sclerostin → formation ON. Romosozumab is an anti-sclerostin antibody that boosts formation.

BOARD TRAP

SOST loss-of-function = sclerosteosis/van Buchem (dense bone). Romosozumab carries a cardiovascular (MI/stroke) boxed warning — avoid post-recent-MI.

12. FGF23 — phosphate hormone

WHAT YOU SEE

FGF23 phosphate-control icon

WHAT IT ENCODES

FGF23 from osteocytes LOWERS serum phosphate: it inhibits renal proximal-tubule phosphate reabsorption (down NaPi cotransporters) and suppresses 1-alpha-hydroxylase (lowers 1,25-vitamin D).

BOARD TRAP

FGF23 excess (XLH, tumor-induced osteomalacia) = low phosphate + INAPPROPRIATELY low/normal 1,25-D; contrast with classic vit-D deficiency where 1,25 may be preserved by secondary hyperPTH.

13. BMU turnover rate

WHAT YOU SEE

Header: 3 wk resorb / 3 mo form

WHAT IT ENCODES

Skeleton receives ~10% of cardiac output, contains ~18% of body Ca turning over yearly. Each BMU: 3 weeks resorption, 3 months formation. PTH (intermittent) is anabolic; continuous PTH is catabolic.

BOARD TRAP

Intermittent teriparatide BUILDS bone; the SAME hormone given continuously (as in hyperPTH) causes net resorption — the classic PTH paradox.
